

Lecture 5: Fourier Series - Let's Look Inside!

CSE 219: Signals and Linear Systems

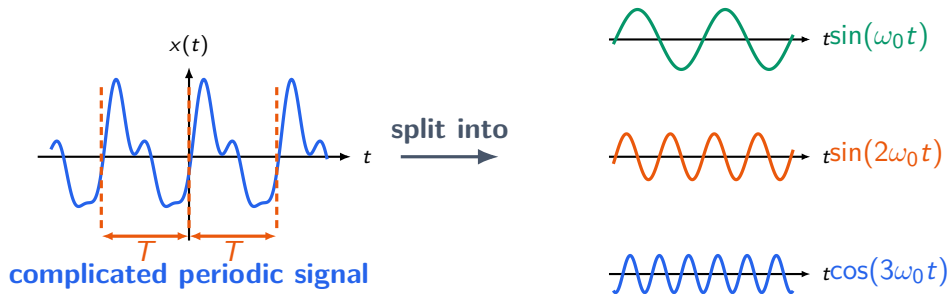
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Fourier Series

Breaking complicated periodic signals into simple waves.

Can we break down a periodic signal into simple sine and cosine waves? Yes, we always can!



complicated periodic signal = sum of simple periodic waves.

Why this idea is useful

1

Some problems are easy for simple waves.

A sine-shaped vibration or temperature pattern.

2

Real initial shapes are usually not simple.

A string can be plucked into a strange shape. A metal rod can have an arbitrary initial heat pattern.

3

Fourier series gives the bridge.

Break the random shape into simple waves, solve each simple wave, then add the answers.

Example: how will a plucked string move?



We want the whole function $u(x, t)$: shape of the string at every position and time.

Please try this visualization on vibrating strings!

Fourier Series and a Vibrating String

Draw any fixed-end initial shape. The app computes its Fourier sine coefficients, evolves each normal mode exactly, and adds the modes back together.

INITIAL SHAPE AND MODEL CONTROLS

Preset

Mode 3

Wave speed c

1.00

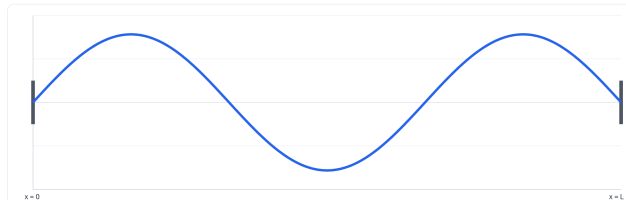
☐ Draw ☒ Initial

Modes N

40

Fixed endpoints are enforced: $u(0,t)=u(L,t)=0$.

STRING MOTION



— $u_N(x,t)$ = Initial shape $f(x)$

▶ Play

⏸ Pause

↺ Reset

t: 0.00 f: 0.50

Turn on **Draw**, then drag on the plot to create a new initial displacement. Releasing from rest means each coefficient is multiplied by $\cos(n\pi ct/L)$.

Another example: how does heat smooth out?



A simple sinusoidal heat pattern is easy to predict. A random heat pattern becomes manageable after breaking it into sinusoids.

Please try this visualization on heat diffusion!

Fourier Series and Heat Diffusion

Draw an initial temperature distribution. The app decomposes it into heat modes, decays each mode analytically, and adds the pieces back together.

INITIAL TEMPERATURE AND MODEL CONTROLS

Preset

Warm left half, cold right half

Thermal diffusivity κ

0.08

Time t

0.00

Modes N

50

Animation speed

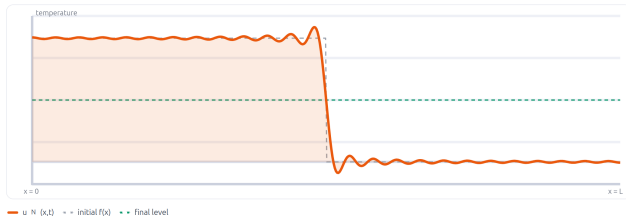
0.20x

Boundary condition

Insulated ends: $u_x(0,t)=u_x(L,t)=0$

☐ Draw ☒ Initial

TEMPERATURE EVOLUTION



The repeated strategy: solve the easy waves, then add

1

Break the initial shape into simple waves.

$$f(x) = \text{wave}_1 + \text{wave}_2 + \text{wave}_3 + \cdots$$

2

Let each simple wave evolve by itself.

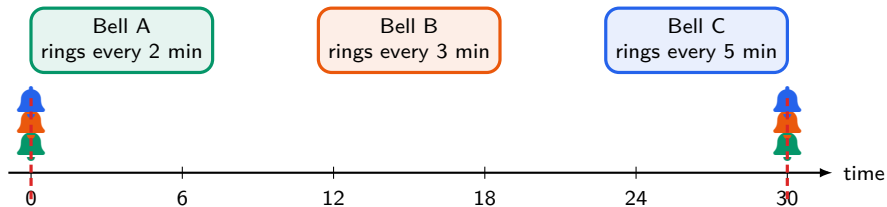
3

Add the evolved waves back together.

The complicated future behavior comes from adding many simple future behaviors.

Bell, bell, bell! Who's there?

A bell analogy



the combined pattern repeats after 30 minutes

The combined period is a common multiple of the individual periods.

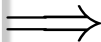
$$\text{lcm}(2, 3, 5) = 30.$$

The same story, but viewed through frequency

Periods

2, 3, 5 minutes

$$T_{\text{combined}} = 30 \text{ minutes}$$



Frequencies

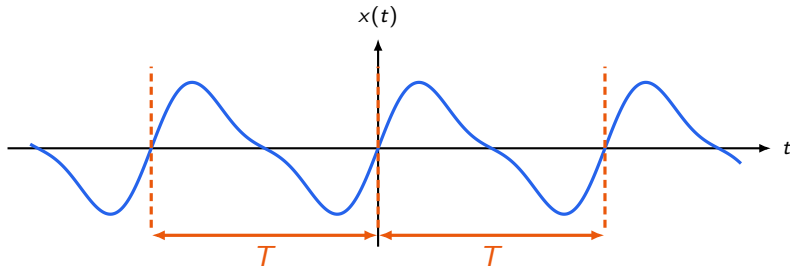
30, 20, 12 times/hour

$$f_0 = 2 \text{ times/hour}$$

The individual frequencies are multiples of the combined frequency:

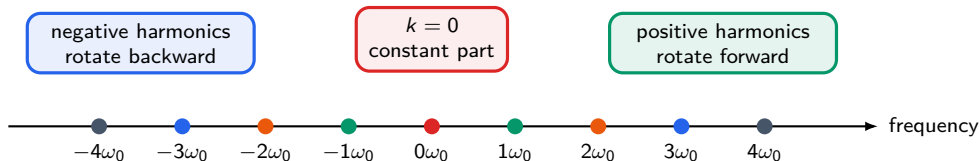
$$30 = 15f_0, \quad 20 = 10f_0, \quad 12 = 6f_0.$$

For a signal with period T , the fundamental frequency is ω_0



$$\omega_0 = \frac{2\pi}{T}$$

Therefore, the allowed building blocks are $k\omega_0$



For a period- T signal, we use frequencies

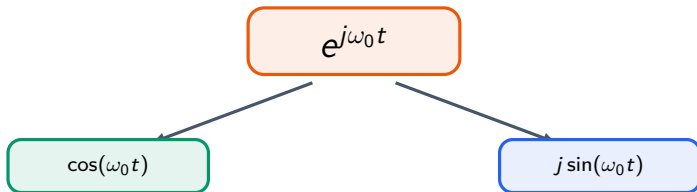
$$\dots, -2\omega_0, -\omega_0, 0, \omega_0, 2\omega_0, \dots$$

or simply $k\omega_0$, where $k \in \mathbb{Z}$.

Instead of sines and cosines, we use complex exponentials

Complex exponentials package sine and cosine together

$$e^{j\omega_0 t} = \cos(\omega_0 t) + j \sin(\omega_0 t)$$



one object carries both parts

So the Fourier series will look like this

Complex exponential Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$e^{jk\omega_0 t}$$

the k -th simple wave

$\times$

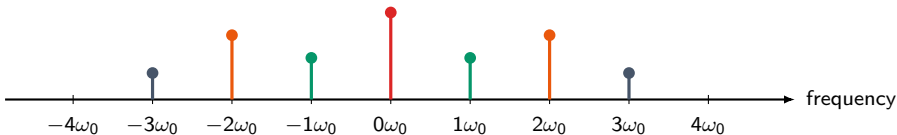
$$a_k$$

how much of that wave

Now the question is: what are the coefficients?

Each coefficient tells us how much of one frequency is present

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$



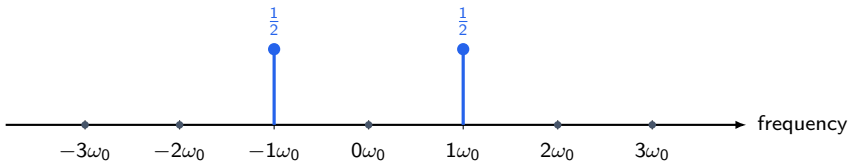
$a_k = 0$: frequency absent

$a_k \neq 0$: frequency present

Example: a cosine already contains two complex waves

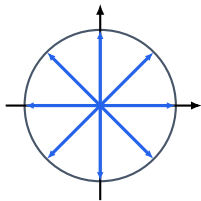
Using Euler's formula

$$\cos(\omega_0 t) = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

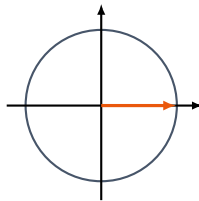


only the $-\omega_0$ and $+\omega_0$ components are present

A small detour: averaging a rotating exponential



rotates
average cancels to 0



does not rotate
average remains 1

$$\frac{1}{T} \int_T e^{jk\omega_0 t} dt = \begin{cases} 1, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

Finding a_0 is easy, just take the average of $x(t)$!

$$x(t) = \cdots + a_{-1}e^{-j\omega_0 t} + a_0 + a_1e^{j\omega_0 t} + a_2e^{j2\omega_0 t} + \cdots$$

$$a_{-1}e^{-j\omega_0 t}$$

↓
average 0

$$a_0$$

↓
survives

$$a_1e^{j\omega_0 t}$$

↓
average 0

$$a_2e^{j2\omega_0 t}$$

↓
average 0

So a_0 is just the average over a period:

$$a_0 = \frac{1}{T} \int_T x(t) dt.$$

How can we find a_1 ? Make the a_1 term stop rotating!

$$x(t) = \cdots + a_{-1}e^{-j\omega_0 t} + a_0 + a_1e^{j\omega_0 t} + a_2e^{j2\omega_0 t} + \cdots$$

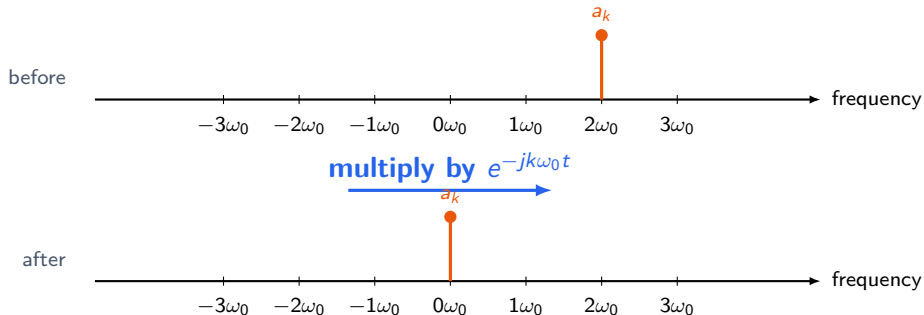
Multiply by $e^{-j\omega_0 t}$

$$x(t)e^{-j\omega_0 t} = \cdots + a_{-1}e^{-j2\omega_0 t} + a_0e^{-j\omega_0 t} + a_1 + a_2e^{j\omega_0 t} + \cdots$$

Now a_1 is non-rotating. So averaging over one period extracts a_1 .

$$a_1 = \frac{1}{T} \int_T x(t)e^{-j\omega_0 t} dt.$$

General idea: move the desired coefficient to zero frequency



Shift a_k to zero frequency, take average. Everything else rotates away.

The coefficient formula

Fourier series coefficient

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

The integral can be taken over any one full period.

$$\omega_0 = \frac{2\pi}{T}, \quad k = 0, \pm 1, \pm 2, \dots$$

Please try building Fourier Series as you wish!

Fourier Series Builder

Build a periodic signal by adjusting its Fourier coefficients. Each coefficient is one rotating complex exponential, and the plotted signal is the real part of their sum.

MODEL CONTROLS

$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{ik\omega_0 t}$, plotting $\text{Re}\{x(t)\}$.

ω_0

period T 2.09

phase θ 0.35

3.0

☐ Keep output real

With the lock on, mirrored coefficients stay equal: $a_{-k} = a_k^*$.

Random mix

Zero signal

Constant

Simple cosine

Stronger cosine

First + second harmonic

Reset

FOURIER COEFFICIENTS

a_0

STATIC

$a_0 = 0.5$



a_1

COUNTERCLOCKWISE

$a_1 = 1.0$



a_{-1}

CLOCKWISE

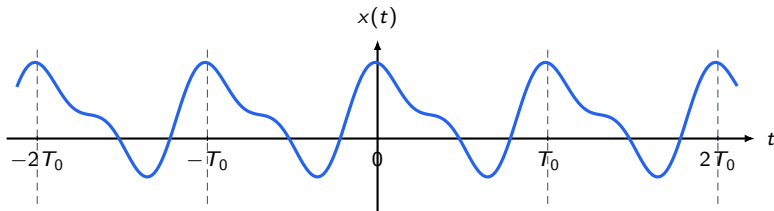
$a_{-1} = 2.0$



Practice Problems on Finding the Fourier Series

Example: can we find the Fourier series without integrating?

$$x(t) = 1 + \sin(\omega_0 t) + 2 \cos(\omega_0 t) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right), \quad T_0 = \frac{2\pi}{\omega_0}.$$



Do we really need to compute $a_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$?

Answer: Yes! The Fourier components are already visible

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}.$$

Components at frequencies $+\omega_0$ and $-\omega_0$

$$\begin{aligned} \sin(\omega_0 t) + 2 \cos(\omega_0 t) &= \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t}) + e^{j\omega_0 t} + e^{-j\omega_0 t} \\ &= \left(1 + \frac{1}{2j}\right) e^{j\omega_0 t} + \left(1 - \frac{1}{2j}\right) e^{-j\omega_0 t}. \end{aligned}$$

The phase-shifted cosine gives the second harmonic

Components at frequencies $+2\omega_0$ and $-2\omega_0$

$$\begin{aligned}\cos\left(2\omega_0 t + \frac{\pi}{4}\right) &= \frac{1}{2}e^{j(2\omega_0 t + \frac{\pi}{4})} + \frac{1}{2}e^{-j(2\omega_0 t + \frac{\pi}{4})} \\ &= \left(\frac{1}{2}e^{j\pi/4}\right)e^{j2\omega_0 t} + \left(\frac{1}{2}e^{-j\pi/4}\right)e^{-j2\omega_0 t}.\end{aligned}$$

Do not forget the constant term

$$1 = 1 \cdot e^{j0\omega_0 t} \quad \implies \quad a_0 = 1.$$

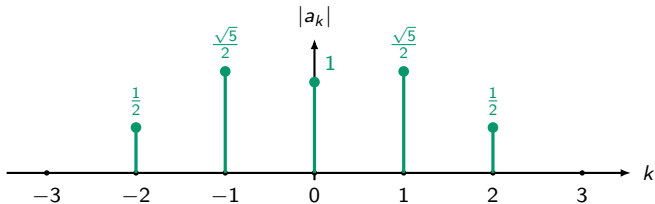
Collect terms with the same rotating frequency

Complex exponential Fourier series

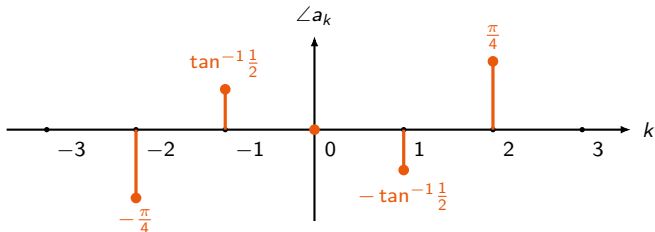
$$x(t) = \frac{\sqrt{2}}{4}(1-j)e^{-j2\omega_0 t} + \left(1 + \frac{j}{2}\right)e^{-j\omega_0 t} + 1 \\ + \left(1 - \frac{j}{2}\right)e^{j\omega_0 t} + \frac{\sqrt{2}}{4}(1+j)e^{j2\omega_0 t}.$$

k	-2	-1	0	1	2
a_k	$\frac{\sqrt{2}}{4}(1-j)$	$1 + \frac{j}{2}$	1	$1 - \frac{j}{2}$	$\frac{\sqrt{2}}{4}(1+j)$

Magnitude spectrum: $|a_k|$

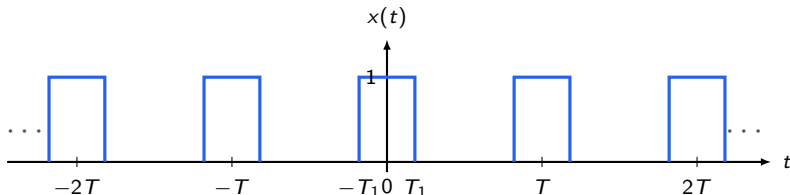


Phase spectrum: $\angle a_k$



Practice: Fourier series of a periodic square wave

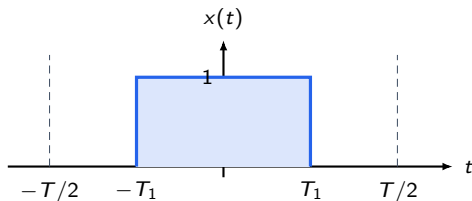
$$x(t) = \begin{cases} 1, & |t| < T_1, \\ 0, & T_1 < |t| < T/2, \end{cases} \quad x(t+T) = x(t), \quad \omega_0 = \frac{2\pi}{T}.$$



Find the complex exponential Fourier series coefficients a_k .

Only the nonzero part contributes to the integral

Choose one convenient period: $-T/2 < t < T/2$



$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt.$$

Outside $-T_1 < t < T_1$, the signal is zero.

For $k \neq 0$, evaluate the short integral

Coefficient for $k \neq 0$

$$\begin{aligned} a_k &= \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt \\ &= \frac{1}{T} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-T_1}^{T_1} \\ &= \frac{2 \sin(k\omega_0 T_1)}{T k\omega_0} \\ &= \frac{\sin(k\omega_0 T_1)}{k\pi}. \end{aligned}$$

For $k = 0$, the coefficient is just the average value

The constant coefficient

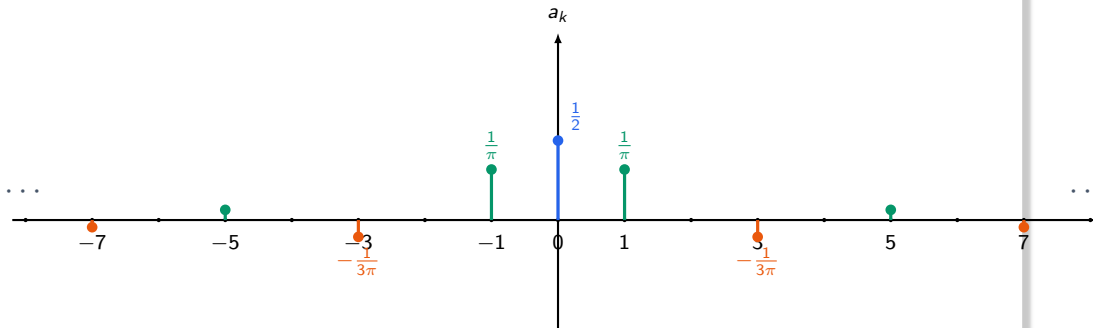
$$a_0 = \frac{1}{T} \int_{-T_1}^{T_1} 1 \, dt = \frac{2T_1}{T}.$$

The average value is just:

$$\frac{\text{time spent at height 1}}{\text{total period}} = \frac{2T_1}{T}.$$

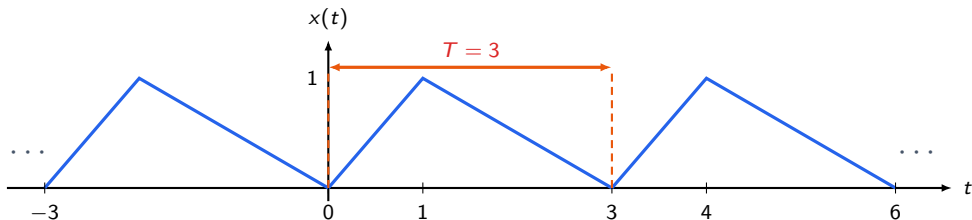
Frequency-domain picture of the square wave, when $T_1 = T/4$

$$a_0 = \frac{1}{2}, \quad a_k = \frac{\sin(k\pi/2)}{k\pi}, \quad k \neq 0.$$



Problem: Fourier series of a repeated triangular ramp

$$x(t) = \begin{cases} t, & 0 \leq t < 1, \\ \frac{3-t}{2}, & 1 \leq t < 3, \end{cases} \quad x(t+3) = x(t).$$



Find the complex exponential Fourier series coefficients a_k .

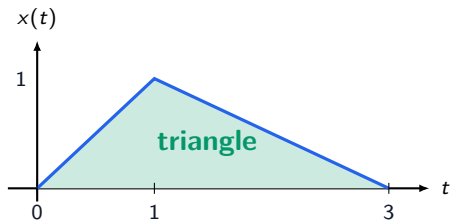
Let's first find a_0 : the average height over a period

$$a_0 = \frac{1}{3} \int_0^3 x(t) dt$$

$$\begin{aligned} a_0 &= \frac{1}{3} \left[\int_0^1 t dt + \int_1^3 \frac{3-t}{2} dt \right] \\ &= \frac{1}{3} \left[\frac{1}{2} + 1 \right] = \boxed{\frac{1}{2}}. \end{aligned}$$

So the average height over one period is $\frac{1}{2}$.

Same a_0 , seen geometrically



$$\text{area over one period} = \frac{1}{2} \cdot 3 \cdot 1 = \frac{3}{2}.$$

$$a_0 = \frac{\text{area over a period}}{\text{period}} = \frac{\frac{3}{2}}{3} = \boxed{\frac{1}{2}}.$$

For $k \neq 0$, split the coefficient integral

$$a_k = \frac{1}{3} \int_0^3 x(t) e^{-jk\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{3}.$$

$$a_k = \frac{1}{3} \left[\int_0^1 t e^{-jk\omega_0 t} dt + \int_1^3 \frac{3-t}{2} e^{-jk\omega_0 t} dt \right].$$

Evaluate the two pieces

For $k \neq 0$,

$$\int_0^1 t e^{-jk\omega_0 t} dt = -\frac{e^{-jk\omega_0}}{jk\omega_0} + \frac{1 - e^{-jk\omega_0}}{(jk\omega_0)^2}.$$

$$\int_1^3 \frac{3-t}{2} e^{-jk\omega_0 t} dt = \frac{e^{-jk\omega_0}}{jk\omega_0} + \frac{e^{-jk\omega_0} - e^{-j3k\omega_0}}{2(jk\omega_0)^2}.$$

$$\text{Since } \omega_0 = \frac{2\pi}{3}, \quad e^{-j3k\omega_0} = e^{-j2\pi k} = 1.$$

The final simplification

The $\frac{1}{jk\omega_0}$ terms cancel:

$$\begin{aligned} a_k &= \frac{1}{3} \left[\frac{1 - e^{-jk\omega_0}}{(jk\omega_0)^2} + \frac{e^{-jk\omega_0} - 1}{2(jk\omega_0)^2} \right] \\ &= \frac{1}{3} \cdot \frac{1 - e^{-jk\omega_0}}{2(jk\omega_0)^2}. \end{aligned}$$

$$\text{Since } (jk\omega_0)^2 = -k^2\omega_0^2,$$

$$a_k = \frac{e^{-jk\omega_0} - 1}{6k^2\omega_0^2}.$$

Fourier series coefficients

$$a_0 = \frac{1}{2}$$

$$a_k = \frac{e^{-jk\omega_0} - 1}{6k^2\omega_0^2}, \quad k \neq 0, \quad \omega_0 = \frac{2\pi}{3}.$$

Please try this visualization on Fourier Series!

🌀 Fourier Series: Building a Wave with Rotating Vectors

Each complex Fourier coefficient becomes one rotating arrow. Add the arrows tip-to-tail, and their endpoint reconstructs the periodic signal.

SIGNAL AND EPICYCLE CONTROLS

Signal type

Sawtooth

Harmonics K

Animation speed

15

Fundamental rate

0.35x

☒ Circles

☒ Vectors

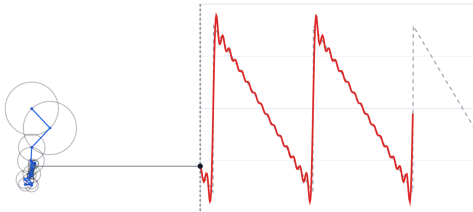
☒ Wave

1.00

Period is normalized to $T = 1$.

ROTATING VECTORS AND WAVEFORM

epicycle chain



Please also try drawing any 2D closed curve!

Fourier Epicycles: Drawing with Rotating Vectors

A closed 2D curve is a complex signal $z(t)=x(t)+jy(t)$. Fourier coefficients become rotating arrows whose endpoint redraws the curve.

DRAWING AND EPICYCLE CONTROLS

Preset shape

Heart

Epicycles

80

☒ Circles ☒ Vectors ☒ Trace ☒ Original

Animation speed

0.35x

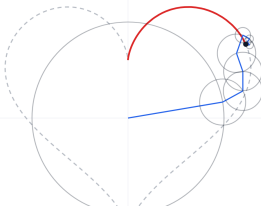
☒ Close curve

Coefficient ordering

Largest first

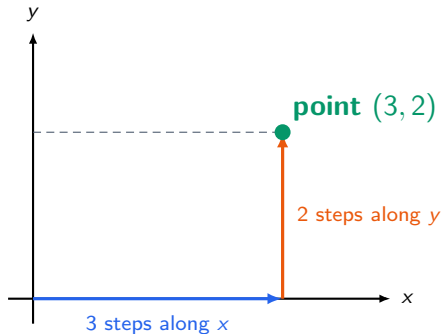
EPICYCLE DRAWING

complex plane



***A beautiful intuition!**

*A coordinate system turns a point into numbers



How much of each direction do we need?

*Functions can also be presented with coordinates

Example: all functions of the form $f(t) = a + bt$

$$f(t) = a \cdot 1 + b \cdot t$$

basis functions

1, t

coordinates

(a, b)

$$3 + 2t \quad \Longleftrightarrow \quad (3, 2)$$

*More axes let us describe more complicated functions

$$f(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots$$

Now the axes are

$$1, \quad t, \quad t^2, \quad t^3, \quad \dots$$

The coordinates of $f(t)$ are

$$(a_0, a_1, a_2, a_3, \dots).$$

***Even e^t has coordinates in this system**

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \dots$$

So in the coordinate system

$$1, t, t^2, t^3, \dots$$

we may think of e^t as having coordinates

$$\left(1, 1, \frac{1}{2!}, \frac{1}{3!}, \dots\right).$$

*Fourier series simply chooses a different coordinate system

Polynomial axes

$$1, \quad t, \quad t^2, \quad t^3, \dots$$

$$f(t) = a_0 + a_1 t + a_2 t^2 + \dots$$

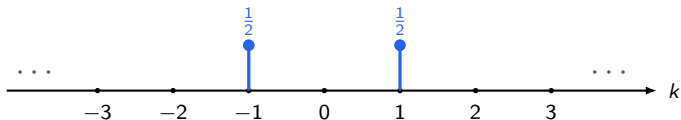
Fourier axes

$$\dots, e^{-j\omega_0 t}, 1, e^{j\omega_0 t}, e^{j2\omega_0 t}, \dots$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

***Example: where is $\cos(\omega_0 t)$ in Fourier coordinates?**

$$\cos(\omega_0 t) = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

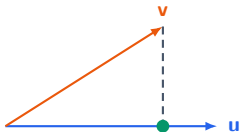


Only $k = -1$ and $k = 1$ are present.
All other Fourier coordinates are zero.

*import projection from linear_algebra

Vectors

coordinate along $\mathbf{u} = \mathbf{v} \cdot \mathbf{u}$



Signals

coordinate along $e^{jk\omega_0 t} = \langle x(t), e^{jk\omega_0 t} \rangle$

projection $\implies$ Fourier coefficient a_k

Same idea: find how much of one axis is present.

***For signals, dot product becomes an average over one period**

$$\langle x(t), z(t) \rangle = \frac{1}{T} \int_T x(t) z^*(t) dt$$

For the k -th Fourier axis,

$$z(t) = e^{jk\omega_0 t} \quad \Rightarrow \quad z^*(t) = e^{-jk\omega_0 t}.$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

Thank you!